

Vehicle-integrated photovoltaics reshape spatiotemporal charging demand and infrastructure needs: An agent-based analysis

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Vehicle-integrated photovoltaics (VIPV) use photovoltaic (PV) modules integrated into an electric vehicle (EV), represented here as rooftop PV, to supply solar electricity directly to the traction battery while driving or parked, reducing reliance on external charging.

Introduction

- Global EV electricity demand reached approximately 250 TWh in 2025 and could exceed 1500 TWh by 2035.
- Treating VIPV as a uniform demand reduction overlooks how battery SoC changes the magnitude, timing, and location of charging events.

Research contributions

- Seasonal VIPV generation is integrated into the established MATSim-UrbanEV framework and evaluated across various scenarios.
- VIPV-modified SoC is propagated through heterogeneous charging decisions to spatially and temporally resolved demand and charger-PV-BESS planning outcomes.

Application for the case of Gothenburg city, Sweden

Population: 557220 synthetic persons; 211880 car users; 50% EV assignment (105940 EV users).

Simulation: 10% sample, continuous seven-day horizon, scaled once to the study population.

Scenarios: noVIPV plus 20%, 50%, and 80% adoption crossed with 400, 700, and 1000 Wp.

Seasonality: spring, summer, autumn, and winter- PV potential profiles

Controlled assignment: vehicle IDs are fixed across capacities and adoption sets are nested.

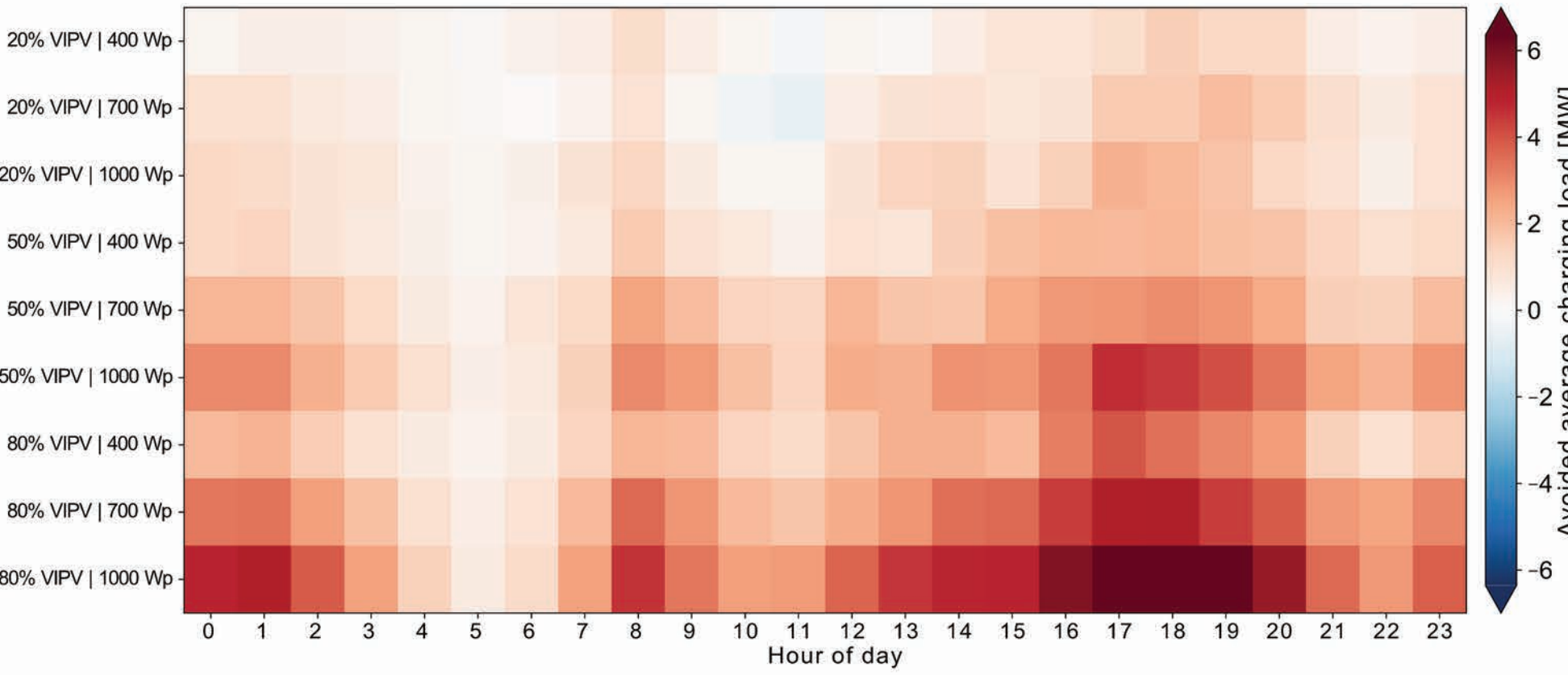
Solar exposure: driving is exposed; parking episodes use the common scenario assumption ($p_{open}=0.5$).

Relief extends beyond solar-generation hours

Seasonally weighted 24-h total and home external-charging profiles

The noVIPV total peak is 41.82 MW at 17:00; the home peak is 30.35 MW at 18:00. Under VIPV80–1000 Wp, maximum hourly relief reaches 6.96 MW at 18:00.

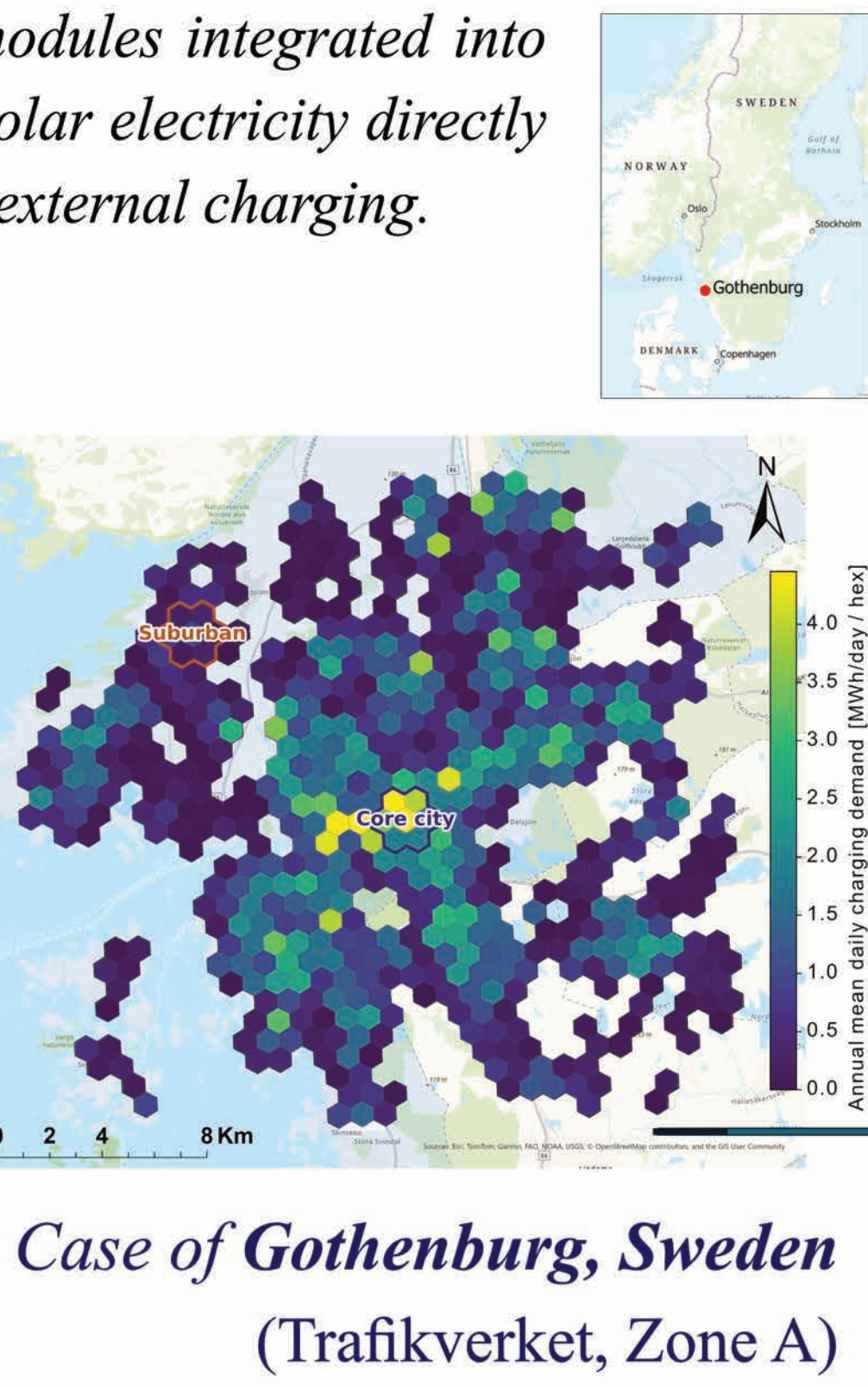
Hourly relief by adoption-capacity scenario



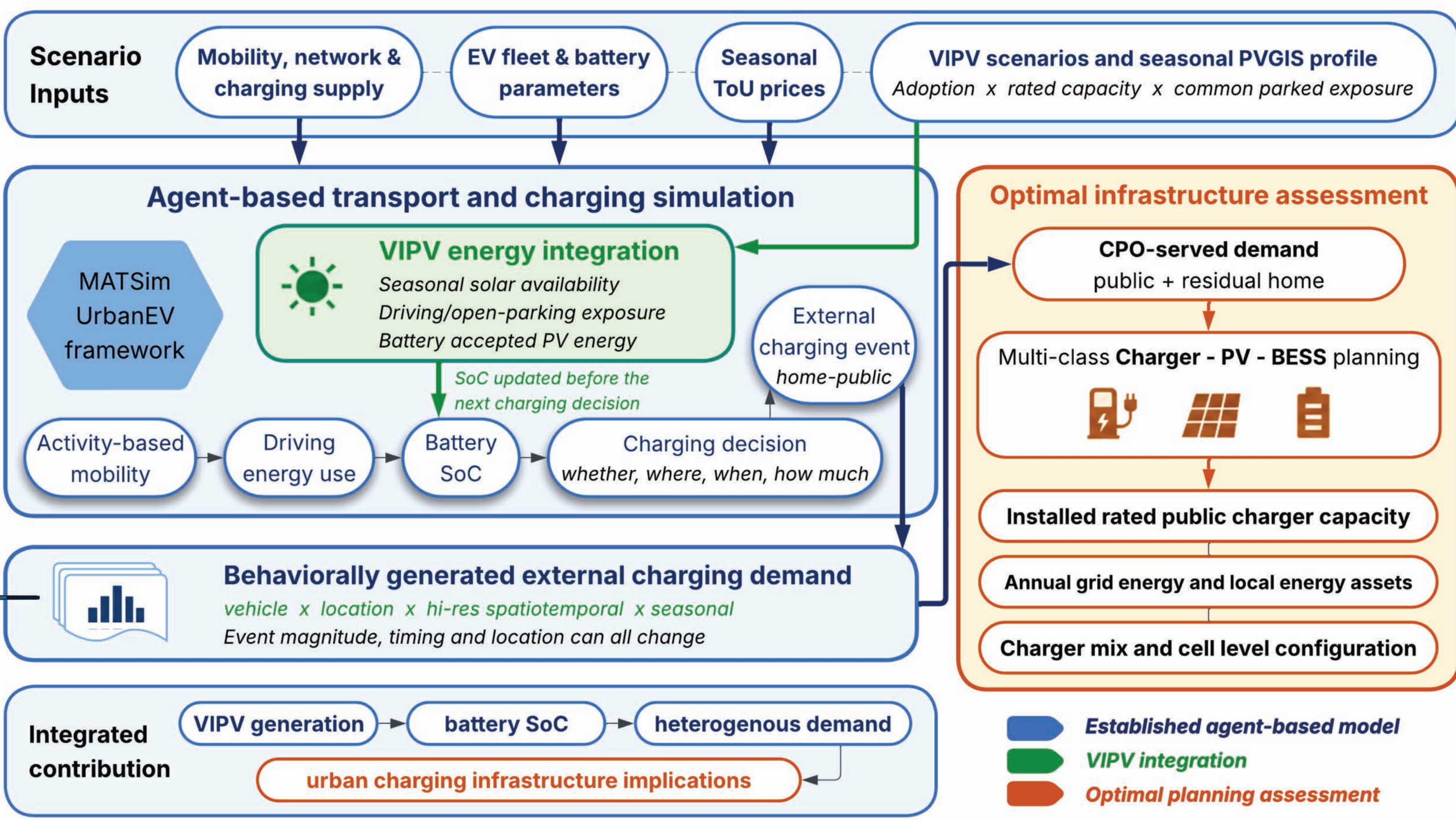
Citywide demand relief is spatially heterogeneous

Annual external-charging relief across the adoption-capacity scenarios. Positive values denote avoided charging, negative values denote local increases.

The noVIPV reference uses 208.59 GWh/yr. Avoided demand rises from 4.72 GWh/yr (VIPV20–400 Wp) to 15.91 GWh/yr (VIPV50–700 Wp) and 34.55 GWh/yr (VIPV80–1000 Wp).



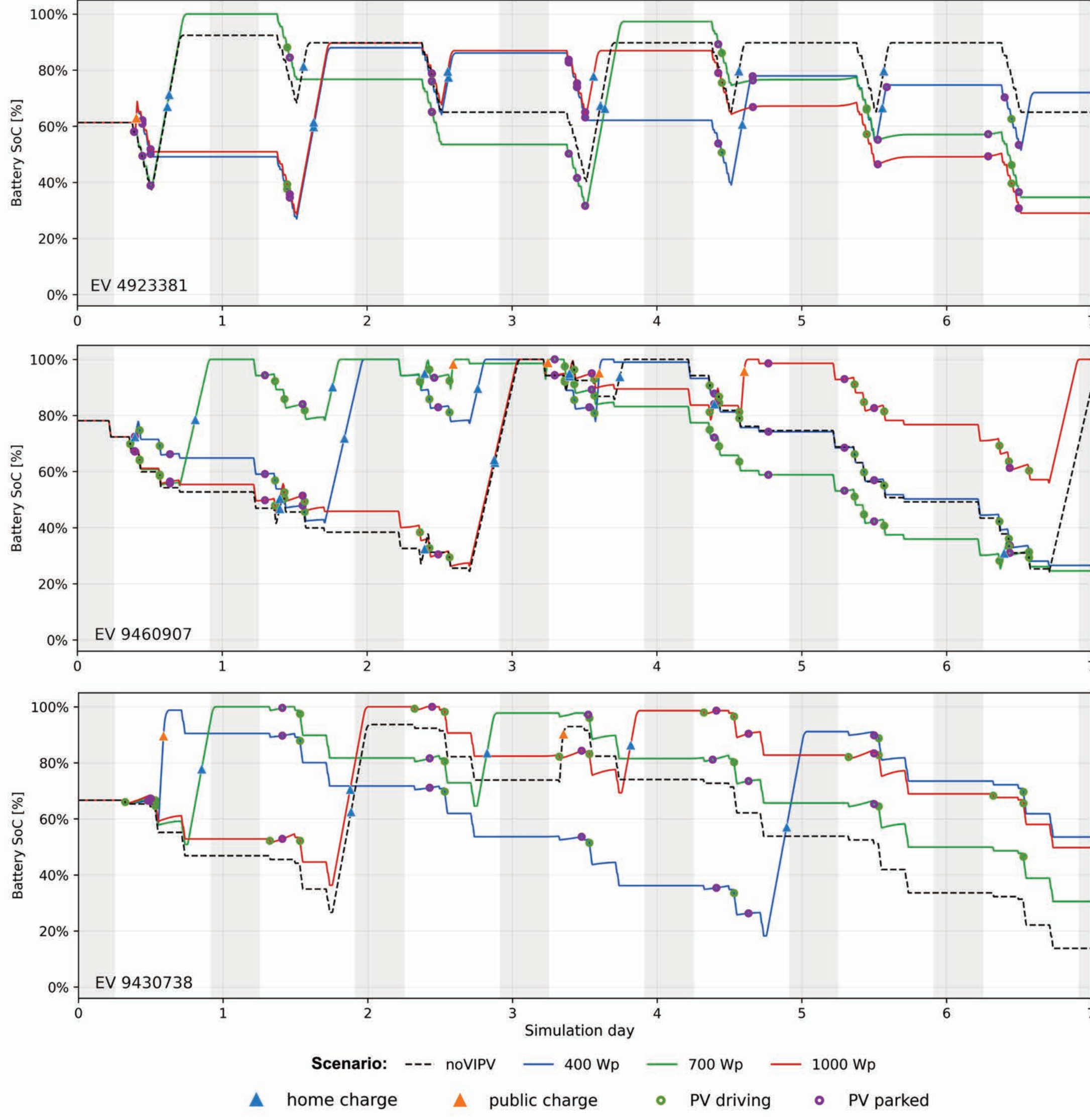
Case of Gothenburg, Sweden
(Trafikverket, Zone A)



The VIPV module supplies battery-accepted onboard energy before the inherited UrbanEV charging decision. Resulting events are retained by vehicle, location, 30-min interval, charging context, and season before entering the downstream planning boundary.

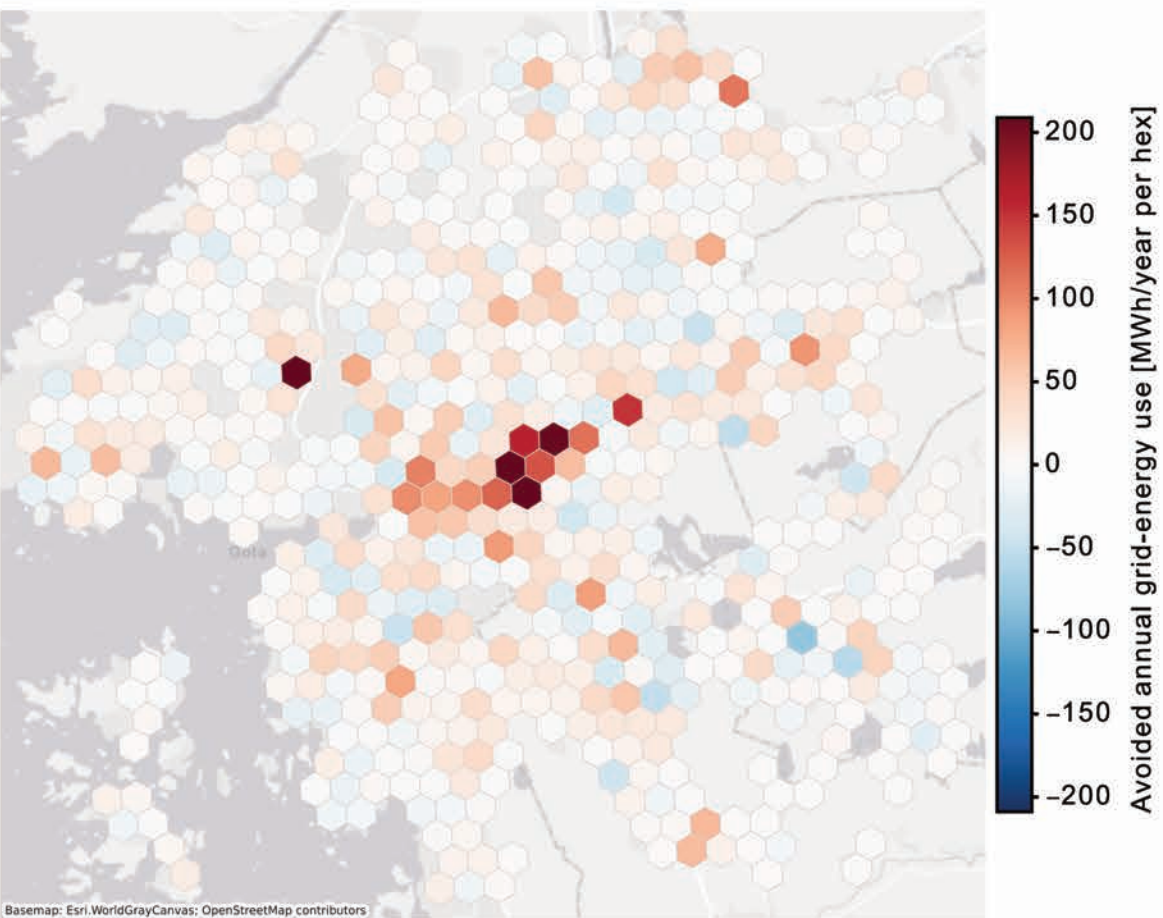
Vehicle-level SoC mechanism: user behavioral impacts

Matched seven-day SoC trajectories for three representative EVs under noVIPV and 400/700/1000 Wp.



VIPV raises SoC between plug-in opportunities, but the realized response depends on each activity chain. Charging can be reduced, delayed, relocated, or retained.

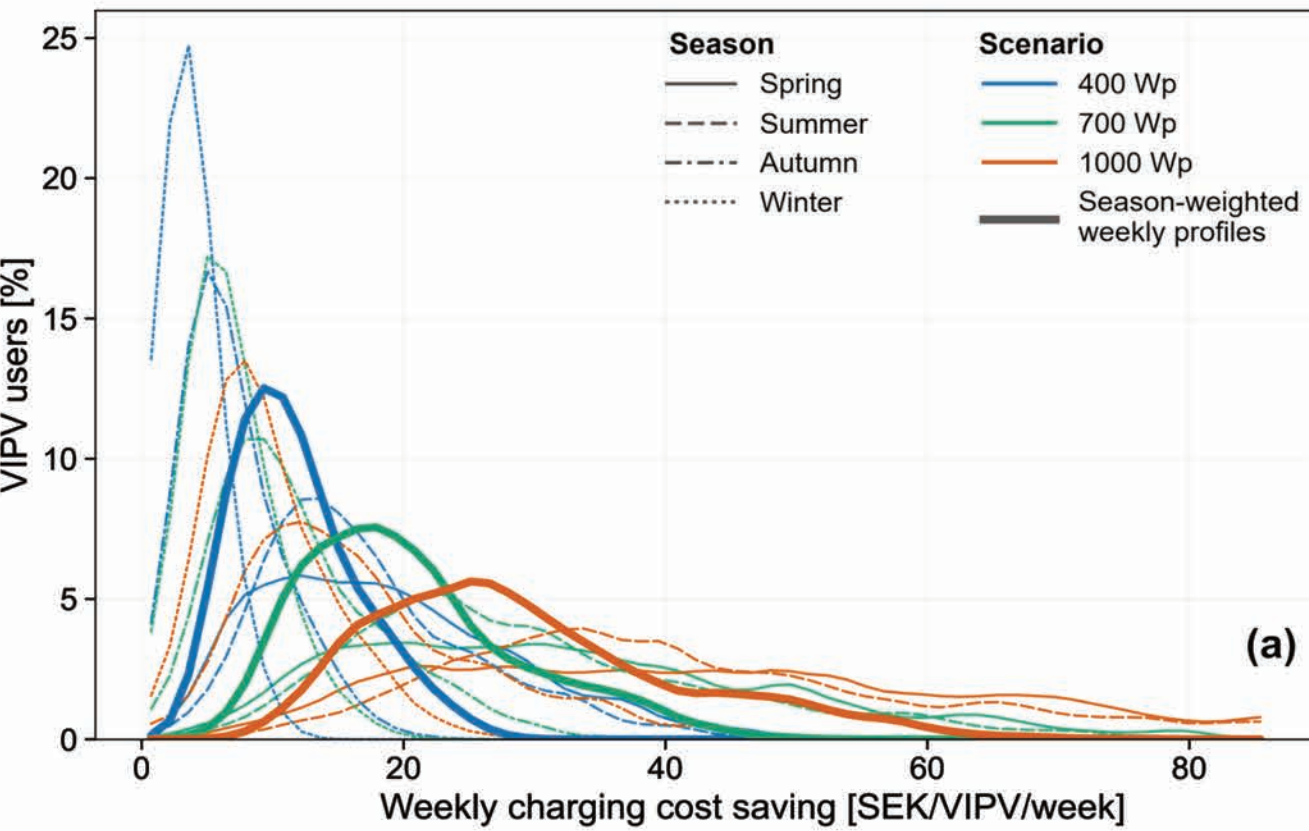
Spatially avoided grid energy use:



Grid use decreases in 339 cells but increases in 192.

Cell-level change from noVIPV to VIPV50_700 Wp in installed rated public charger power and annual grid-energy use.

Weekly avoided charging-cost by season for VIPV users



Useful stored PV is attributed first to a fixed mobility-energy requirement and valued using each vehicle's noVIPV cost and recorded-duration intensities.

Downstream optimal infrastructure reconfiguration

Optimized annual charging infrastructure, energy, and CPO economic outcomes for the demand scenarios.

Metric	Unit	noVIPV	VIPV50–700 Wp	Change
Charging and local-energy infrastructure				
Slow chargers	count	606	617	+1.8%
Medium chargers	count	2 608	2 418	-7.3%
Fast chargers	count	165	163	-1.2%
Installed public charger capacity	kWh/30 min	36 146	34 067	-5.8%
Stationary PV panels	count	69 810	63 649	-8.8%
BESS units (10 kWh)	count	2 295	2 132	-7.1%
Annual energy flows				
Grid electricity purchased	GWh/yr	67.50	61.88	-8.3%
PV electricity used locally	GWh/yr	28.68	25.94	-9.5%
BESS discharge	GWh/yr	9.56	8.91	-6.8%
Annual CPO economic terms				
Public charging revenue	MSEK/yr	585.28	534.70	-8.6%
Grid electricity cost	MSEK/yr	115.56	106.08	-8.2%
Annualized charger CapEx	MSEK/yr	24.08	22.99	-4.5%
Annualized PV+BESS CapEx	MSEK/yr	35.34	32.36	-8.4%
Net profit	MSEK/yr	410.29	373.25	-9.0%

Change is calculated relative to noVIPV. Stationary PV uses 500 Wp panels. All planning parameters are identical between the two optimizations.

Key take-aways

- VIPV alters charging behavior rather than applying a uniform energy offset. The share of vehicles showing no reduction or an increase in external charging fell from 29.6% at 400 Wp to 20.0% at 700 Wp and 13.0% at 1000 Wp.
- At 700 Wp, VIPV reduces the seasonally weighted weekly charging-cost equivalent by 13.9% (20.60 SEK per vehicle) and the recorded charging-duration equivalent by 14.6% (0.84 h per vehicle).
- VIPV is not a uniform percentage subtraction from charging energy. It changes battery states before behavioral charging decisions, allowing event magnitude, timing, and location to change.